Readme of scripts and data used in "*Seascape genomics reveals candidate molecular targets of heat stress adaptation in three coral species*"

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This repository contains scripts and data to replicate the analysis of the paper. In this file the content of each folder is overviewed, and a description for every script and data employed.

Note: In this scripts, AM=A. millepora, PDA= P. damicornis alpha (=P. damicornis in the paper) and PDB=P. damicornis beta (=P. acuta in the paper).

Folder "CommonScriptsAndData":

→ functions.R : contains custom R functions that are required during the various pipelines.

→ sampling_sites_env.csv: contains the values for every environmental variable at every site.

→ site-sample.csv: contains information on the origin (i.e. sampling site) of every sample.

Folder "ScriptsAndData":

A) Folder "ScriptsAndData/Scripts":

Contains all the scripts to run the seascape genomics analysis on one of the three species. The scripts are sorted in the order of execution:

→ **1-FilterGeneticData.R :** filters genotype matrix for missing values, clones, LD.

→ **2-CheckCorrelationEnvVars.R :** checks correlation between environmental data

→ **3-LFMM.R :** launches the LFMM analysis.

→ **4-ProcessOutputLFMM.R** : Postprocessing of LFMM output.

→ **5-Annotate_significant_SNPs.R :** Retrieve closest genes to significant SNPs found with LFMM.

→ **6-PrepareSetRank.R :** Prepare the analysis of gene ontology enrichment.

→ **7-RunSetRank.R :** Run GO enrichment analysis of genes next to SNPs associated with heat stress.

→ **8-OutlierTest.R :** Genome scan to detect outlier loci.

→ **9-PopStrAnalysis.R :** Analysis of population structure.

→ **10-MAFbySite.R :** Evaluate variation in minor allele frequency between sampling locations.

Folders "AMData", "PDAData" and "PDBData":

These folders contain each input files necessary to run the pipeline of the Scripts folder for each of the three species (AMData=A. millepora, PDAData=P. damicornis and PDBData=P. acuta). NB: the .rda files can be opened using the "load" R function.

The files are:

- **env.rda:** environmental data per sample.
- **exons.rda:** table recording the position of every exon annotated.
- **fSNPS.rda:** filtered genotype matrix.
- **GENpos.rda:** table recording the position and annotations of every gene annotated.
- **groupvar.rda:** list of environmental variable highly correlated (0.9).
- **outlierloci.rda:** output of the outlier test analysis.
- **outlist.rda:** output of LFMM, formatted for post-processing.
- **reef.rda:** identity of sampling site for every sample.
- **Setrank***:** set of output files of the SetRank analysis.
- **snpINFO.rda:** contains the position of every SNP across the genome.
- **sorted_imputed.vcf:** the genotype matrix before filtering.
- **QS.CRWBA.rda:** list of q-value for all the SNPs vs. BAF (5 km).
- **QS.CRWBA_1.rda:** list of q-value for all the SNPs vs. BAF (1 km).
- **ranked_CRWBA.rda :** list of genes in proximity to SNPs associated with heat stress, ranked by q-value (BAF 5 km).
- **ranked_CRWBA_1.rda:** list of genes in proximity to SNPs associated with heat stress, ranked by q-value (BAF 1 km).
- **SRcollection.rda:** gene set collection of GO molecular functions for SetRank.